1. Which of the following best describes data communication?
   A. The process of teaching in a classroom
   B. The exchange of data between devices through some form of transmission medium
   C. The process of storing data only
   D. The creation of a lesson plan

2. In a data communication system, the *sender* is:
   A. The device that receives the message
   B. The device that generates and sends the data
   C. The medium through which data travels
   D. The rules for communication

3. Which of the following is not a component of data communication?
   A. Sender
   B. Receiver
   C. Medium
   D. Curriculum

4. The rules that govern data communication are known as:
   A. Protocols
   B. Topologies
   C. Algorithms
   D. Manuals

5. Which signal type varies continuously over time?
   A. Digital signal
   B. Analog signal
   C. Binary signal
   D. None of the above

6. Which of the following is a digital communication advantage over analog?
   A. More affected by noise
   B. Easier to process and store
   C. Lower security
   D. Requires no conversion

7. The physical path between sender and receiver is called:
   A. Protocol
   B. Medium
   C. Topology
   D. Node

8. Which topology connects all devices to a single central hub?
   A. Bus
   B. Star
   C. Ring
   D. Mesh

9. The Internet is an example of:
   A. LAN
   B. WAN
   C. MAN
   D. PAN

10. A LAN is typically used for:
   A. Connecting computers within a school building
   B. Connecting countries
   C. Linking cities
   D. Satellite communication

11. Which of the following is a wireless communication technology?
   A. Ethernet
   B. Fiber optics
   C. Wi-Fi
   D. Coaxial cable

12. Which device connects two different networks and directs data between them?
   A. Switch
   B. Router
   C. Hub
   D. Repeater

13. A protocol used for sending emails is:
   A. HTTP
   B. SMTP
   C. FTP
   D. POP

14. Which protocol is used for secure web communication?
   A. HTTP
   B. HTTPS
   C. FTP
   D. SNMP

15. The unit used to measure network speed is:
   A. Hertz
   B. Bits per second[bps]
   C. Joules
   D. Bytes only

16. In networking, IP stands for:
   A. Internet Protocol
   B. Internal Program
   C. Instant Process
   D. Integrated Platform

17. Which of the following is not an example of a transmission medium?
    A. Copper cable
    B. Fiber optic cable
    C. Radio waves
    D. Keyboard

18. In a teacher's office network, the computers share files and printers. This is an example of:
    A. Data processing
    B. Network resource sharing
    C. Wireless broadcasting
    D. Signal conversion

19. In data communication, noise refers to:
    A. Background music
    B. Unwanted signals that interfere with communication
    C. Classroom discussion
    D. Teacher's instructions

20. Which layer of the OSI model is responsible for data delivery from one node to another?
    A. Physical
    B. Data Link
    C. Network
    D. Application

21. The main purpose of data encoding is:
    A. To make lessons interesting
    B. To convert data into a suitable format for transmission
    C. To store files
    D. To speed up typing

22. Which transmission mode sends data in both directions, but one at a time?
    A. Simplex
    B. Half-duplex
    C. Full-duplex
    D. Multiplex

23. Which network device regenerates signals to extend transmission distance?
    A. Switch
    B. Repeater
    C. Router
    D. Modem

24. In networking, bandwidth refers to:
    A. The physical width of the cable
    B. The amount of data that can be transmitted per second
    C. The brand of network cables
    D. The number of connected devices

25. The main benefit of networking for teachers is:

A. Reducing lesson planning time

B. Sharing resources and information easily

C. Making classrooms smaller

D. Eliminating exams

SECTION B

[75 Marks]

Answer any three questions in this section

1 a. What is data communication?*[5 marks]*

b. List and explain the five[5] main components of a data communication system with examples from a school environment.*[15 marks]*

c. State two differences between a sender and a receiver in data communication.*[5 marks]*

2 a. What is a network topology? *[5 marks]*

b. Describe three (3) types of network topologies (star, bus, ring) and give one advantage and one disadvantage of each. *[15 marks]*

c. Differentiate between a hub and a switch. *[5 marks]*

3a. What is data flow in networking? [*5 marks]*

b. Explain the three (3) types of data flows with examples. *[15 marks]*

c. Differentiate between analog and digital signals. *[5 marks]*

4a. Define data communication in the context of education. *[5 marks]*

b. Explain five[5] ways data communication can enhance teaching and learning. *[15 marks]*

c. State five benefits of using wireless networks in schools. *[5 marks]*

5a. What is the OSI model? *[5 marks]*

b. List and explain the seven (7) layers of the OSI model with one example function for each. *[15 marks]*

c. Differentiate between the physical layer and the application layer. *[5 marks]*